DO NOW — RECALL

Lesson 122 · Selecting a Development Method: Case Stu

Name: _____ Class: _____ Date: _____

1. What Boolean operator joins invalid cases score<0 and score>100?

2. A linked-list node has data and what other essential field?

3. In a BST, where is a value less than the current node placed?

4. Why can an array provide direct indexed access?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

OR.

2. Model answer

A link/reference to the next node (for a singly linked list).

3. Model answer

In its left subtree (under the chosen no-duplicate convention).

4. Model answer

Its element locations can be calculated from base address and index.

By the end, you can…

• Separate stakeholder requirements, constraints, risk and available resources.

• Justify one method against credible alternatives with evidence.

• Write and test one key decision procedure that fits the brief.

Vocabulary — know these terms

functional requirement — Behaviour a system must provide.

non-functional requirement — Quality or constraint on system operation.

acceptance criterion — Condition showing requirement has been met.

trade-off — Choice balancing competing advantages and costs.

NEW LEARNING

Read, annotate and explain

Analyse the brief

Separate stakeholder requirements, constraints, risk and available resources.

Choose a lifecycle

Justify one method against credible alternatives with evidence.

Trace a core algorithm

Write and test one key decision procedure that fits the brief.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A regional bus service needs journey booking with changing fare rules and a fixed launch date. Choose an approach and an algorithm.

Model answer

Use iterative development for changing fare rules with documented acceptance milestones for launch. Fare algorithm: collect zones and passenger category, select valid tariff, compute fare, reject invalid input and test boundary zones.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State two functional and two non-functional requirements for an event-booking service. [6]

Question 2

Justify a suitable lifecycle for a well-understood regulated payroll update. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Design an algorithm for rejecting bookings when capacity is full and explain test cases and lifecycle fit. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The development method should be chosen without reference to users, change risk or deadlines.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.